

Housing Affordability Across Ontario Municipalities

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Introduction and Purpose

Affordability is major concern across Ontario.

- ▶ A recent poll conducted by Angus Reid identified the cost of living and housing affordability as a the most important issue.

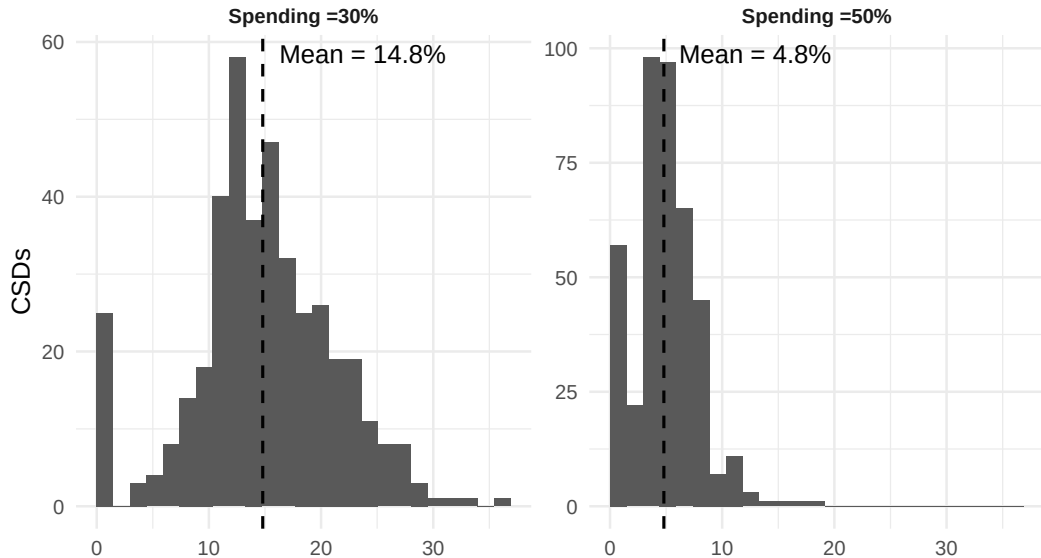
This presentation will seek to **identify which factors are most associated with affordability** using Statistics Canada data and a regression model.

Statistics Canada provided the two main data sources for this presentation at the **Census Subdivision level (CSD)**:

1. Table 98-10-0247-01 provided affordability variables.
2. Table 98-401-X2021021 is the summary of the predictor variables.

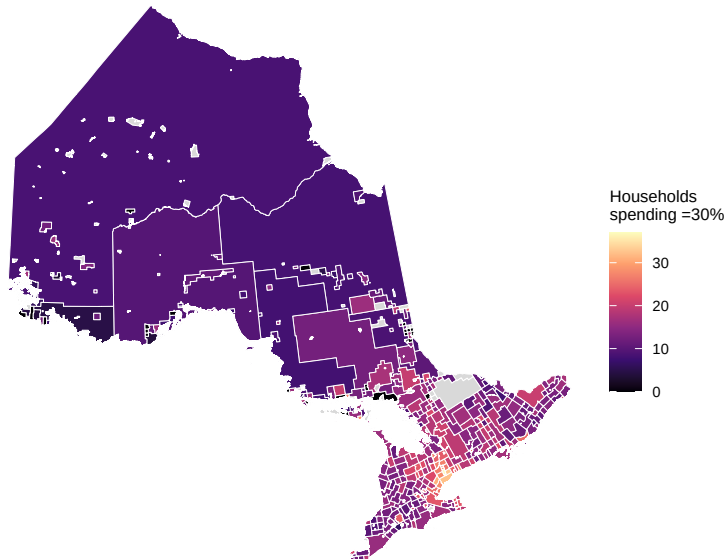
EDA: Affordability Outcome Distribution

Distribution of housing affordability (n=577, missing=168)



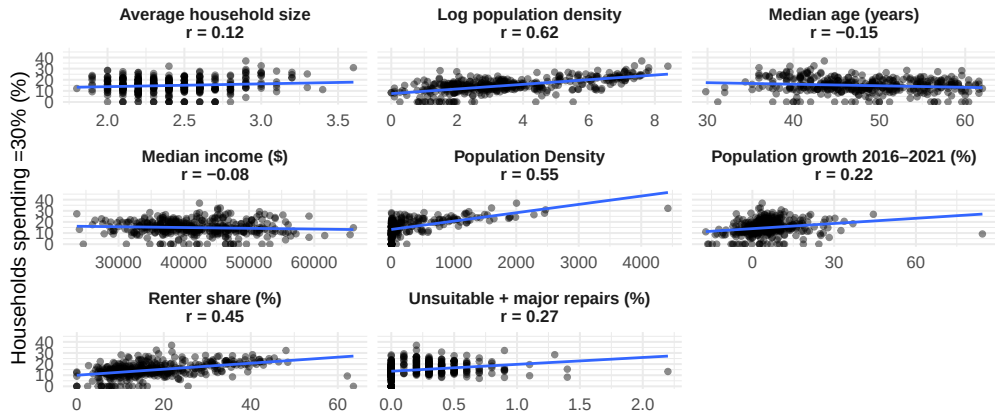
Affordability Map

Housing affordability burden across Ontario



EDA Predictors

Univariate relationships with housing affordability burden



Modelling Approach

Table 1: Summary of candidate predictors

Variable	mean	median	q1	q3
Percent Renting	17.7	15.1	10.0	22.8
Median Age	48.3	48.0	43.2	54.1
Median Total Income	4.2	4.1	3.7	4.6
Hundres of people per km2	2.0	0.2	0.1	0.9
Average Household Size	2.4	2.4	2.2	2.6
Not suitable and major repairs needed	0.3	0.0	0.0	0.3
2016 to 2021 population growth	5.8	4.8	0.3	9.3

We run three types of regression to model the affordability metrics.

1. OLS: modeling the mean predicted value of affordability
2. Fractional Logit: treats affordability as an expected proportion, ensuring all fitted values are between 0 and 1

Best way to model

- ▶ We compared OLS and fractional logistic regression and found very little difference in predictive performance.

Table 2: 10-fold cross-validation RMSE (percentage points)

Outcome	OLS	Fractional logit
30%+ affordability	4.6	4.6
50%+ severe affordability	2.5	2.5

In addition, we conducted sensitivity analyses:

Model Results

Table 3: Fractional logistic regression results

Predictor	30%+ Estimate	30%+ AME	50%+ Estimate	50%+ AME
(Intercept)	0.915***	-74.607	1.238***	-36.437
renter_pct	0.006***	0.301	0.007**	0.093
median_age	0.008***	0.432	0.010***	0.187
income_10k	0.046*	-1.175	0.059	-0.235
density_100	0.008**	0.313	0.009**	0.129
avg_hh_size	0.190***	11.881	0.280***	4.882
unsuitable_repairs_pct	0.084**	3.188	0.080***	1.551
pop_growth_pct	0.004***	0.185	0.007*	0.073

- ▶ As we can see the estimates between the two outcomes are not substantially different.
 - ▶ A one-person increase in average household size is associated with an approximately **12 percentage-point increase** in the share of households experiencing affordability challenges.
 - ▶ A one-point increase in the percent of unsuitable/disrepair dwellings is associated with a **3 percentage-point increase** in the share of households experiencing

Implications and other data sources:

- ▶ The CMHC and Ontario Government provide two additional data sources.

\begin{table} \caption{Ten CSDs with the highest 30%+ affordability burden}

Municipality	Household Size (2.6)	% Renter (22.8%)	30%+ (%)	Housing target status	2020 starts per 1,000
Richmond Hill	2.9	21.8	36.9	Not met	0.0
Markham	3.0	18.3	32.6	Not met	0.0
Toronto	2.4	48.1	32.2	Exceeded	0.0
Brampton	3.6	21.9	30.9	On track	0.7
Aurora	2.9	18.8	29.3	Not met	0.0
Parry Sound	2.0	40.4	28.5		0.0
Tehkummah	2.0	10.2	28.3		0.0
Mississauga	2.9	29.6	27.9	Not met	0.0
Vaughan	3.1	14.1	27.8	Not met	0.0
Oshawa	2.6	35.8	27.5	Not met	0.1

\end{table}

- ▶ Among the ten CSDs with the highest affordability burden, only **Brampton and Oshawa** had more than 0.1 housing starts per 1,000 residents in the available 2020 CMHC data.
- ▶ Moving forward, only the City of Toronto has exceeded its housing target.

Limitations

There are many limitations to this study:

1. There is no time element.
 - 1.1 We're unable to capture the trend in affordability. Using cross sectional data prevents us from answering the very simple question of 'is affordability worse?'.
1.2 CMHC data only went as far back as 2020. Housing starts captures January 2020 starts and does not allow us to see a five or ten-year trend.
2. The 2021 Census was affected by Covid-19

From the regression model, we could conclude that the improving these two policies would improve affordability:

1. Repairing the existing housing stock
2. Funding for more growth through Building Faster Fund